



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Software Engineering

BSCS Section 6E

Project Proposal

Plasma: The Social Intelligence Layer for Steam

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1. Project Title

Plasma: The Social Intelligence Layer for Steam

2. Main Features of the Product

Plasma is designed to bridge the gap between "launching a game" and "living a gaming life." While Steam dominates as a marketplace, the modern PC gamer's library is fragmented across Battle.net, Riot, Epic, and standalone clients. Plasma acts as a specialized "Second Screen" web application that aggregates a user's entire gaming identity, both automatic (Steam) and manual (everything else), into a rich, community-focused social hub.

It utilizes a modern, reactive web interface designed to visualize the social heartbeat of a user's gaming network, repurposing the raw data from Valve's ecosystem into meaningful social interactions.

"The Pulse" (Unified Activity Feed)

A dynamic, algorithm-free timeline where users discover the real-time gaming life of their peers. Unlike the static lists found in standard launchers, The Pulse turns raw data into a visual story.

- **Live Action Feed:** Users can see exactly what their network is doing—not just "Playing Dota 2," but "Ranked Match - 45 mins in."
- **Rich Media Sharing:** Users can post "Moments" directly to the feed (screenshots, clips, or text updates like "Finally beat Malenia!").
- **Deep Linking:** The "Join" button on any friend's activity uses browser protocols (steam://) to instantly launch the game and connect to their lobby.

"The Omni-Library" (Manual Non-Steam Support)

Recognizing that not every game lives on Steam, Plasma offers a robust Manual Integration Protocol. This allows users to build a complete "Shelf" of their gaming history, regardless of where the game was bought.

- **Database Selection:** Users can search a global database of games (e.g., Valorant, League of Legends, Fortnite) and manually add them to their Plasma Library.
- **Status Toggles:** Users can manually toggle their status for these titles (e.g., set status to "Playing Valorant"), which pushes a notification to The Pulse just like a Steam game would.

Intent Modes

To prevent social friction and misalignment, users can toggle their current "Social Intent" on their profile, signaling their availability to the network:

- **Comp Mode:** Highlights a user's rank and "Tryhard" status. Signals they are looking for serious teammates for ranked playlists in games like CS2 or Apex Legends.
- **Chill Mode:** Highlights single-player activity or casual arcade modes. Signals they are open to chat but not looking for intense gameplay.

Secure Connection Protocol (The Lobby System)

To ensure a toxicity-free environment, direct messaging and invites are built on a Double Opt-In basis, similar to the "handshake" protocols in secure networking.

- **Mutual Consent:** Users cannot spam invites to 'unknowns'. A direct chat or invite only goes through if both users follow each other or both users have RSVP'd to the same "Rally" event.
- **Reputation Gating:** Users can set their lobbies to "Verified Only," requiring joining players to have a certain Trust Factor or account age on Steam to prevent smurfing/trolling.

"The Rally" (Event Coordination Engine)

Finding a time to play with a full squad is arguably harder than the games themselves due to time zones and adult responsibilities.

- **Visual Scheduling:** Plasma solves this with a "Rally," a calendar-based event system. A user creates a Rally (e.g., "Friday Night Raid - 9 PM EST"), and the system automatically converts that time to every friend's local time zone.
- **Role Call:** Users RSVP with specific roles (e.g., "I'll play Support," "I'll be the AWPPer"). This prevents the "who is playing what?" chaos before the game starts.
- **Smart Pings:** Since Plasma is a web app, it sends browser push notifications 15 minutes before the Rally starts, alerting users even if they aren't at their PC.

"The Prestige" (Meta-Gamification)

A dedicated feature for the "Achievement Hunters." This is a gamified profile section that aggregates a user's entire Steam history into a single "Gamer Score."

- **Rarity Weighting:** Unlike Steam, which treats all achievements equally, Plasma weighs them by global rarity. Unlocking a 0.1% achievement grants significantly more "Plasma XP" than a 50% one.
- **The Hall of Fame:** Users can pin their proudest accomplishments to the top of their profile, creating a curated "resume" of their gaming skills (e.g., displaying a "Global Elite" rank next to a "100% Completion" badge for Elden Ring).
- **Hybrid Tracking:** For Steam games, progress is automatic. For Non-Steam games, users can manually "Claim" major milestones (e.g., "Reached Diamond in LoL") which are then verified by peer endorsements or screenshot uploads.

Verified Steam Identity

Access is strictly limited to users who link a valid SteamID64 via OpenID authentication.

- **Anti-Smurf:** This ensures a high-trust environment where every profile corresponds to a legitimate main account, filtering out throwaway hacker accounts or bots.

3. Expected User Groups

We've identified four core personas within the PC Gaming ecosystem that Plasma serves:

- **The Shotcallers (Squad Leaders):** The organizers of the friend group. They are the ones constantly asking "When are we playing?" and "Who is online?" They need "The Rally" to schedule sessions and "The Pulse" to see who is available without messaging five different people.
- **The Completionists (Achievement Hunters):** High-achievers who play games to 100% completion. They care about badges, rarity, and statistics. They use The Prestige to display their rare trophies and compare "Gamer Scores" on the leaderboard, which Steam does not natively offer in a competitive format.
- **The Variety Gamers (The Socialites):** Players who jump between trending games (e.g., Palworld, Helldivers) solely to hang out with friends. They use Intent Modes to signal they are just "Chilling" and rely on the feed to see what the group is playing this week.
- **The Pro-Am Aspirants (Competitive Players):** Players focused on climbing ranks in competitive shooters or MOBA titles. They use Comp Mode to find reliable teammates and filter out casual players who might hurt their rank.

4. Domain Knowledge

To ensure success, we are leveraging knowledge from the following domains:

Industry & Market Trends

The Unbundling of Discord: While Discord is the king of chat, it is becoming "noisy." Users are seeking specialized spaces for organizing play rather than just talking over each other. Plasma separates the planning from the chatting.

Asynchronous Socializing: Adult gamers often cannot play every day. Plasma allows them to stay connected via The Pulse (viewing friends' achievements and updates) even when they don't have time to log into a game themselves.

HCI & Social Computing

Contextual Awareness: The interface changes based on user intent. If a user is in "Comp Mode," the UI prioritizes stats and win rates. If in "Social Mode," it prioritizes screenshots and status updates.

Cognitive Load Reduction: Organizing a 5-stack raid usually involves checking Discord, Steam Chat, and WhatsApp. Plasma centralizes this into one Rally dashboard, reducing the mental effort required to start a game session.

Variable Reward Schedule: The "Pulse" feed utilizes variable rewards (seeing a friend's rare drop or high score) to keep users engaged and checking the app daily.

Gaming Psychology

The "Third Place" (Digital): Plasma serves as the digital lobby—a space that is neither "in-game" (high

stress) nor "real life" (boring). It is a neutral ground for banter and planning.

Social Validation: Gamers crave recognition for their skills. The Prestige system leverages the psychological need for status by turning invisible Steam data into a public, competitive leaderboard.

FOMO (Fear Of Missing Out): By broadcasting upcoming Rally events on the main feed, Plasma triggers a positive FOMO loop, encouraging users to RSVP so they aren't left out of the session.

5. Competitive Analysis

While general social networks and chat apps exist, none specifically cater to the "Meta-Layer" of the Steam ecosystem with this level of utility.

Competitor	Strengths	Weaknesses	How Plasma Wins
Steam Community	Native integration; user base of millions.	Cluttered, outdated UI (2010s era). The activity feed is hidden and rarely checked. Chat is basic.	Modern UX: Plasma extracts the data but presents it in a responsive, "Twitter-style" feed that prioritizes engagement over store sales.
Discord	The gold standard for voice chat and communities.	Fragmented "Server" structure makes it hard to see a global view of what all friends are doing. No structured event scheduling.	Unified Layer: You don't "join" a server; you see your entire network. The Rally offers structured scheduling that Discord's scrolling chat lacks.
Reddit (r/gaming)	Great for news and anonymous discussion.	Anonymous and public. You cannot coordinate a game session with trusted peers easily.	Verified Identity: Plasma is a "Walled Garden" for your actual Steam friends, not anonymous strangers.
Twitter / X	Good for following developers/influencers.	Too much noise (politics, ads). Not integrated with game data (can't see stats/ranks).	Game-Native Data: Plasma is purely gaming. Every post is context-aware, linking directly to games, stats, or achievements via the Steam API.
Epic Games Store	Exclusive titles and free games.	Zero social features. No profiles, no forums, no activity feeds. A "storefront only" experience.	The Social Bridge: Plasma provides the social layer that platforms like Epic completely lack, though we focus on Steam for the richest data integration.